Abhinav Nandwani

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EDUCATION

University of Wisconsin-Madison — Madison, WI
B.S Computer Engineering and Computer Science (Double Major) — *Expected May 2027*
GPA: 3.9/4.0

EXPERIENCE

AMD — Boxborough, MA
Applied AI Engineer — *August 2025 - Present*

- Bootstrapped AMD's internal AI benchmarking function, building evaluation harnesses for agentic systems (Claude Code, GPT-5 Codex, Gemini, and internal agents) on silicon design tasks — spec-to-RTL, waveform debug, design verification — using verifiable rewards via UVM testbenches and formal equivalence checks; results directly inform agent improvement cycles.
- Built agentic waveform debug tool **deployed to 1,000+ engineers** across AMD's chip design teams, cutting debug cycles from 2 weeks to 2 days; developed waveform MCP enabling agents to trace signals, analyze waveforms, and debug simulation runs directly.
- Extended benchmarking expertise to evaluate external AI vendors and startups building tools for chip design, directly informing AMD leadership on adoption and acquisition decisions.

SiliconBadgers | siliconbadgers.com — Madison, WI
Chief Technical Advisor — *Present*

- Lead technical direction for UW-Madison's **120+ member tapeout club**, guiding undergraduate, master's, and PhD students across RTL, software, verification, and physical design.
- Drive **hardware/software co-design for LLM prefill, decode, and disaggregated inference**, developing custom **end-to-end quantization schemes** targeting state-of-the-art accuracy and efficiency with fully quantized execution.
- Define system interfaces, **SystemVerilog coding standards, and verification requirements**; lead architecture and code reviews and establish CI workflows to integrate subteam contributions toward FPGA prototyping and ASIC tapeout.

Professor Daifeng Wang Lab — Madison, WI
Undergraduate Researcher — *January 2025 - August 2025*

- Built inference and fine-tuning pipelines for single-cell genomics foundation models processing 4M+ profiles from PsychAD Consortium dataset (2.5M cells, 3 modalities), reducing adaptation time by 40%.
- Developed phenotype classification models achieving 90%+ accuracy with clustering and trajectory inference discovering 5+ novel cell-state transitions in Alzheimer's progression.
- Built distributed training infrastructure on CHTC GPU Lab across 10+ NVIDIA H200 GPUs with Docker/Singularity workflows, reducing deployment overhead by 60%.
- Optimized multi-GPU fine-tuning with PyTorch DDP achieving 3.2x faster convergence. Scaled PEFT (LoRA/QLoRA) from 4 to 32 GPUs, reducing costs 75%.

PROJECTS

Xilinx FPGA AI Inference Accelerator — GitHub

- Architected hardware-accelerated LLM inference engine on AMD Xilinx ZYNQ 7020 FPGA. Designed 32x32 INT8 systolic array with dataflow optimization executing llama.cpp at silicon level with PS-PL heterogeneous integration via Vitis HLS and Vivado.
- Engineered memory hierarchy with BRAM banking and double-buffering reducing DRAM bandwidth 85%, achieving 120 GOPS throughput at <500mW power for edge inference.
- Demonstrated 6x speedup and 40% efficiency gains over ARM Cortex-A9. Validated via ILA/VIO profiling, establishing FPGA as viable GPU alternative for edge AI.

Pipelined RISC CPU — GitHub

- Designed 5-stage pipelined RISC processor in Verilog with custom ISA, hazard detection, and forwarding logic. Architected datapath achieving single-cycle throughput for non-dependent instructions.
- Implemented 2-bit dynamic branch predictor achieving 85% accuracy. Reduced pipeline stalls by 60% through speculative execution and branch target buffering.
- Engineered LRU cache hierarchy with 16KB 4-way I-cache and 32KB 8-way D-cache. Implemented write-back and write-allocate policies reducing memory latency 75%.
- Synthesized to 1 GHz on Synopsys 32nm library with $<5mm^2$ area. Achieved 100% functional coverage through constrained random verification.

TECHNICAL SKILLS

Languages: Verilog, SystemVerilog, Python, C/C++, Bash, Java
ML/AI Frameworks: PyTorch, TensorFlow, CUDA, NumPy, scikit-learn
Tools: Altera, Questa, Vivado, Vitis, Synopsys DC, Git, Linux, Embedded C, HTCondor